Rocks and minerals

The outer layers of Earth are mostly solid rock. This is easy to see where there are mountains or canyons, but much more rock is hidden under the soil and the sea. Rocks are made of minerals. They can be changed or destroyed by weather or water at the surface, or by heat and pressure inside Earth.

WHAT'S THE DIFFERENCE?

Minerals are natural chemical substances that usually form as solid crystals. Each type can be recognized by its hardness, colour, and atomic structure. Rocks are a mixture of minerals locked together. For example, granite is made of the minerals quartz, feldspar, and mica.


MINERAL: QUARTZ

IGNEOUS ROCKS

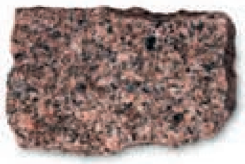
Formed from volcanic material as it cools down, igneous rocks are of two types. Some, such as pegmatites, form deep underground. Others, such as andesite, form when volcanic lavas cool at Earth's surface.


PUMICE


ANDESITE


TOURMALINE PEGMATITE


OBSIDIAN


PINK GRANODIORITE


RHYOLITE


PORPHYRY

SEDIMENTARY ROCKS

These rocks form mostly at the bottom of seas and lakes. They are made from grains of sand and clay worn away from older rocks by wind and water. Over a very long period of time, the grains settle into layers of mud, or sediment. These layers are buried and eventually harden into new rock.


GREYWACKE


LIMESTONE WITH FOSSILS


FLINT


OIL SHALE


SHALE


PUDDINGSTONE

MINERALS

There are thousands of different minerals, though only about 30 make up most rocks. They usually form from water solutions and molten rock, sometimes deep inside Earth. Some, such as diamonds, are cut and polished to make gemstones.


CHRYSOPRASE


MAGNETITE


ENARGITE


TENNANTITE


COCKSCOMB BARITE


ARSENOPYRITE


HORNBLENDE


BROOKITE


CHALCOCITE


HALITE


CHALCOPYRITE


NAILHEAD CALCITE


GALENA


CORUNDUM


SULPHUR


GOLD